

Python Control Statements – Interview Questions

Topics: if, if-else, if-elif-else, nested if, for, while, range(), break, continue, pass, and loop else.

- Q1. What are control statements in Python? Why are they used?
- Q2. What is the purpose of the if statement in Python?
- Q3. What is the difference between if and if-else statements?
- Q4. Explain the syntax of an if-else statement with a simple example.
- Q5. What is an if-elif-else statement? When would you use it?
- Q6. Can an if statement exist without an else statement? Explain.
- Q7. What happens if none of the conditions in an if-elif-else statement are True?
- Q8. Can we use multiple elif blocks in Python? Explain with an example.
- Q9. What is nested if? When is it useful?
- Q10. What is the difference between nested if and if-elif-else?
- Q11. What is the purpose of the for loop in Python?
- Q12. What is the purpose of the while loop in Python?
- Q13. What is the difference between for and while loops?
- Q14. How does the range() function work with a for loop?
- Q15. What is the difference between range(5), range(1, 5), and range(1, 10, 2)?
- Q16. What happens if the condition of a while loop is always True?
- Q17. What is an infinite loop? How can you stop or avoid one?
- Q18. What is the purpose of the break statement?
- Q19. What is the purpose of the continue statement?
- Q20. What is the difference between break and continue?
- Q21. What is the purpose of the pass statement?
- Q22. What is the difference between pass and continue?
- Q23. Can break be used outside a loop? What happens if we try?
- Q24. Can continue be used outside a loop? What happens if we try?
- Q25. How does break behave when used inside nested loops?
- Q26. How does continue behave inside a nested loop?
- Q27. What will happen if an else block is used with a for loop?
- Q28. What is loop else in Python, and when does the else block execute?

- **Q29.** Write a program using control statements to check whether a number is positive, negative, or zero.
- **Q30.** Write a program using a loop and control statements to print numbers from 1 to 20 while skipping even numbers and stopping after 15.